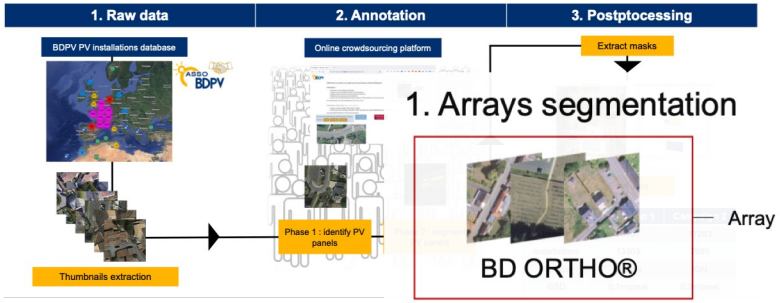
# How to certify a deep learning-based solar arrays detection model?

## Crowdsource training data



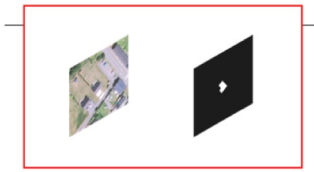

1. Arrays segmentation



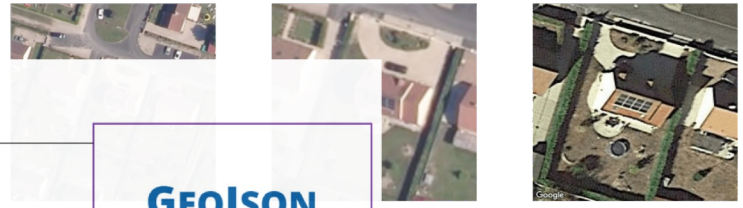

Array ?

Yes

No



## Understand sensitivity to image shifts



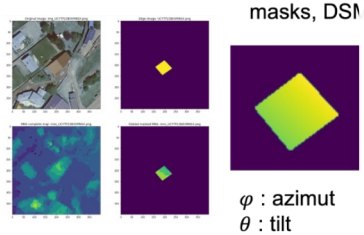
GEOJSON

(left : IGN, right : Google).
fully account for variations in image quality

Classification | Segmentation | Conversion

## Robustly extract characteristics

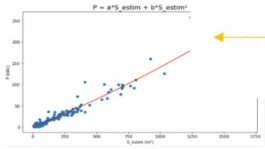
Extraction of the tilt angle and a zimut using segmentation masks, DSM and Theil-Sen estimator



$\varphi$ : azimut
$\theta$ : tilt

Plots : Y. Tremenbert

Polynomial fit betw en the installed capacity and surface



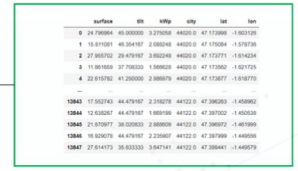
2. Characteristics extraction and filtering

GEOJSON

BDPV data

BD TOPO®

Characteristics extraction | Filtering | PV registry

## Monitor outputs' accuracy

Average percentage error (APE) of the estimated re-aggregated installed capacity compared to the installed capacity per city recorded in the RNI